

Arduino for Musicians

Alex Grant

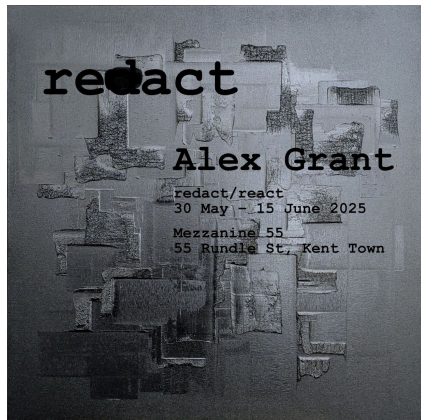
Shameless Plug

redact/react

30 May - 15 June

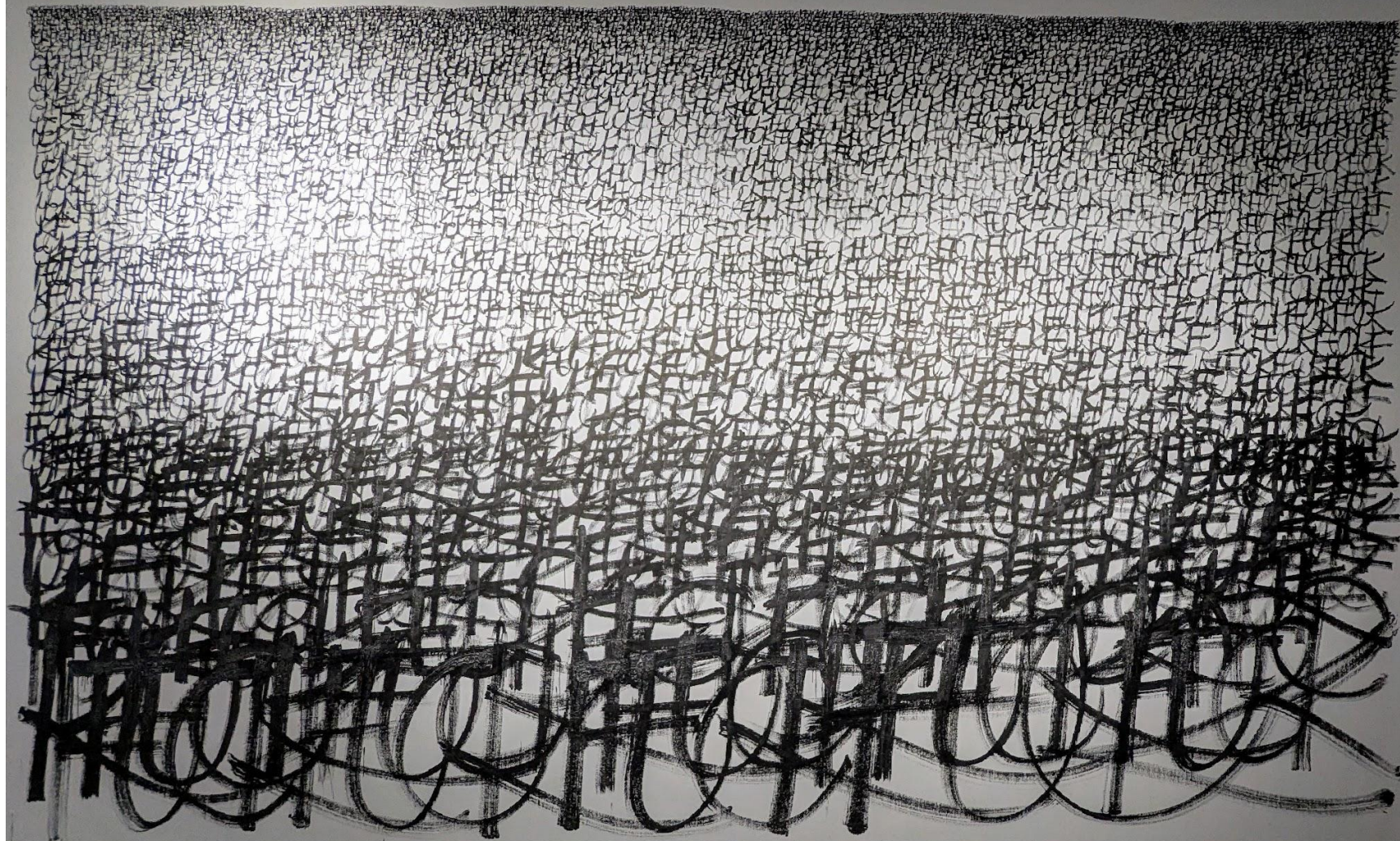
Mezzanine 55

Solo exhibition



Opening 6:00 - 20:00 Friday 30 May

Artist talk 2pm Saturday 7 June



Workshop Plan

Week 1 [Tues, May 13th]

Introduction to Arduino: Focusing on the Arduino microcontroller, participants will be introduced to basics of electronics.



Week 2 [Tues, May 20th]

MIDI: An introduction to the MIDI standard for interconnecting music equipment. Participants will learn how MIDI sends messages for notes and control values. We will build a simple system using the Arduino to send MIDI commands to a digital audio workstation.

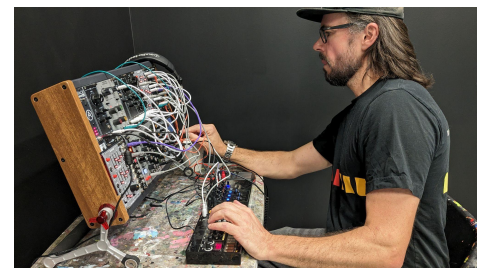


Week 3 [Tues, May 27th]

Modular Synthesis: Using VCV Rack participants will explore sound design and music production using modular synthesis.

Week 4 [Tues, June 3rd]

Homebrew MIDI Controllers: Bringing together the previous three workshops, in this session, participants will build custom MIDI controllers using Arduino and various sensors.



Prerequisites

Week 1 [Tues, May 13th]

- Laptop with [Arduino IDE](#) installed
- Optional: [fritzing](#)
- USB-A to USB-C converter or hub if needed
- Arduino starter kit (provided)



Week 2 [Tues, May 20th]

- Laptop with [Ableton Live](#) and [MIDIView](#) installed
- Alternatively [Garageband](#), [Tracktion](#) or whatever DAW
- Headphones
- Arduino starter kit



Week 3 [Tues, May 27th]

- Laptop with [VCV Rack](#) Installed
- Headphones



Week 4 [Tues, June 3rd]

- Laptop with Arduino IDE, Ableton Live, VCV Rack and MIDIView installed
- Headphones
- Arduino starter kit



Lab Guest
TheLab2024?

Download the Code



<https://github.com/grantaj/labrats-arduino>

git is a tool for tracking changes in files

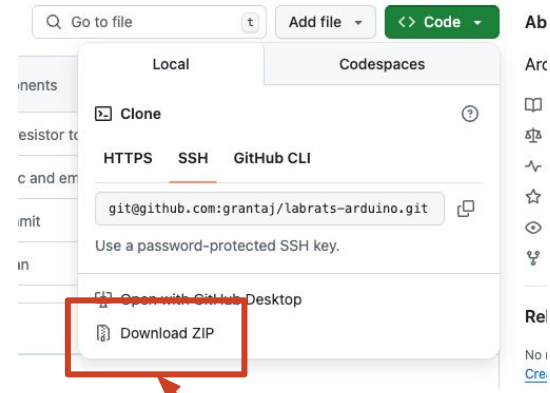
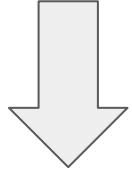
Saves versions of work so you can go back to earlier stages

You can work on different features or ideas in parallel using "branches"

Git works locally, but you can also use websites like **GitHub** to share your work and collaborate

Helps everyone stay organized and avoid overwriting each other's changes

Every change you save with Git is called a "commit", each commit has a message describing the change.



download the code as a .zip archive to your computer

Sounds Design using Modular Synthesis

Outline

What is Modular Synthesis?

Voltage controlled oscillators

Envelope generators

Voltage controlled filters

Low frequency oscillators

Modulation

Sound design for “drums”

What is Modular Synthesis?

- Modular synthesis builds custom instruments using separate **sound-making** and **sound-shaping** components called **modules**.
- Modules have specific functions:
 - Generating sound (e.g. oscillators)
 - shaping it (e.g. filters and envelopes),
 - Controlling other modules (e.g. LFOs, sequencers).
- Modules are connected using patch cables —there's no fixed layout.
- It's hands-on and experimental

Audio Signals

Represent sound waves—what you actually hear.

Typically range from 20 Hz to 20,000 Hz (the human hearing range).

Produced by modules like oscillators, and flow through modules like filters, and VCAs to shape the sound.

Patched to outputs (like speakers or headphones) or to other audio-processing modules.

Control Voltage (CV)

Signals representing **instructions (controls)**, not sound—control other modules.

Range from -5V to +5V or 0V to +10V, depending on the system.

Can control pitch, volume, filter cutoff, modulation depth, and more.

Types of CV sources include LFOs, envelopes, sequencers, and manual knobs.

CV lets you animate or automate your patch—e.g., sweep a filter, trigger a drum, or change pitch over time.

Automatically turning knobs



**modular is
addictive and
expensive**

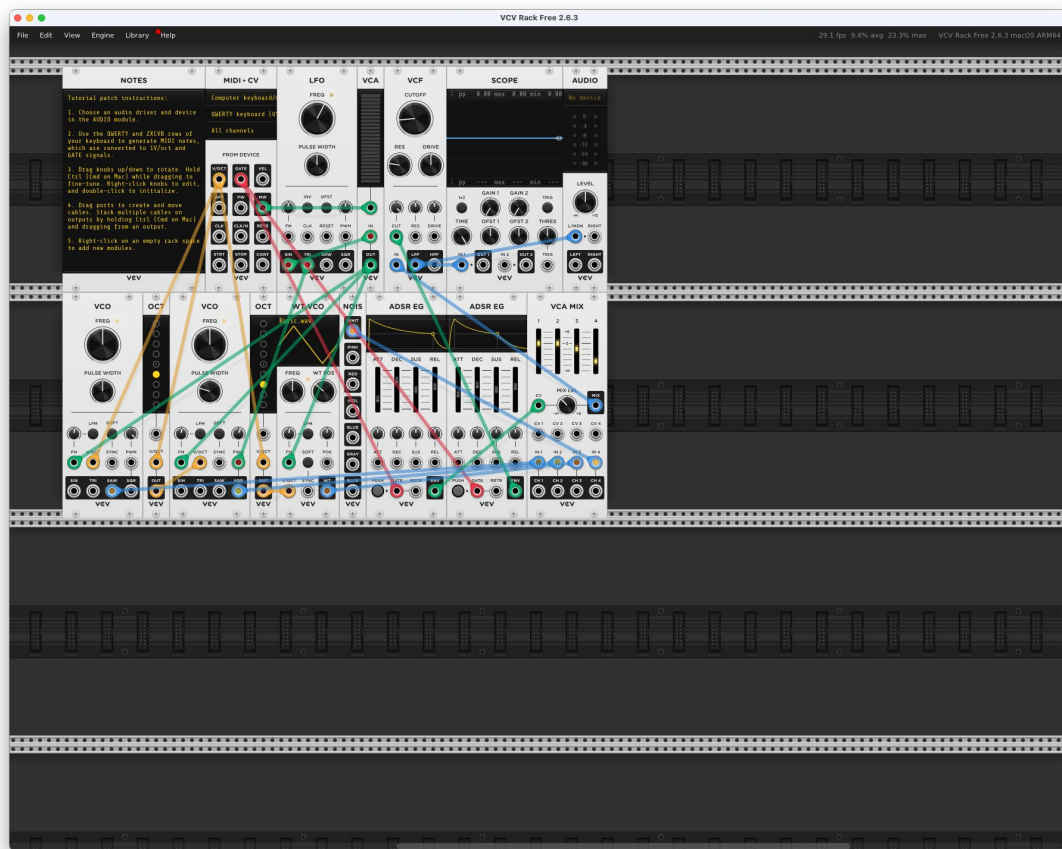
VCV Rack



<https://vcvrack.com/>

Free! (but also paid modules)

Many software versions of
hardware modules



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Voltage Controlled Oscillators

Week-3-Modular/patches/1-VCO.vcv

Configure Audio output for your computer

Adjust FREQ

Change oscillator type: SIN/TRI/SAW/SQR

Scope shows the waveform



Add a spectrum analyzer

- Got to vcvrack.com and Log in (you will need an account to access modules)
- Go to the VCV Library <https://library.vcvrack.com/Bogaudio>
- Click the “+ Subscribe” button next to Bogaudio
- Launch VCV Rack on your computer.
- “Library” menu → “Update all”
- Once installed & relaunched:
 - Right-click on empty rack space to open the module browser.
 - Search for "Analyzer".
 - Select Bogaudio Analyzer and click to add it to your rack.
- Patch **SCOPE OUT1** to **ANALYZER IN & ANALYZER THRU** to **L/MON**

Spectrum analyzer

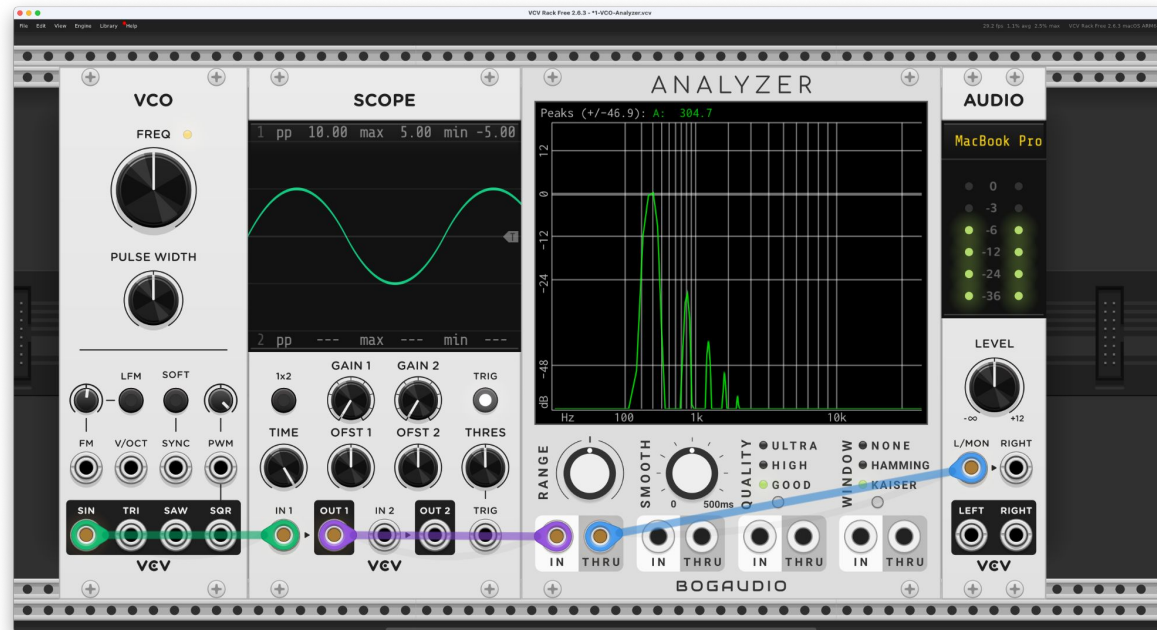
2-VCO-Analyzer.vcv

Shows the frequency content

Patch each of the VCO waveforms
(SIN/TRI/SAW/SQR) to ANALYZER INs

The different peaks are **harmonics**

These give the **timbre** of the waveform



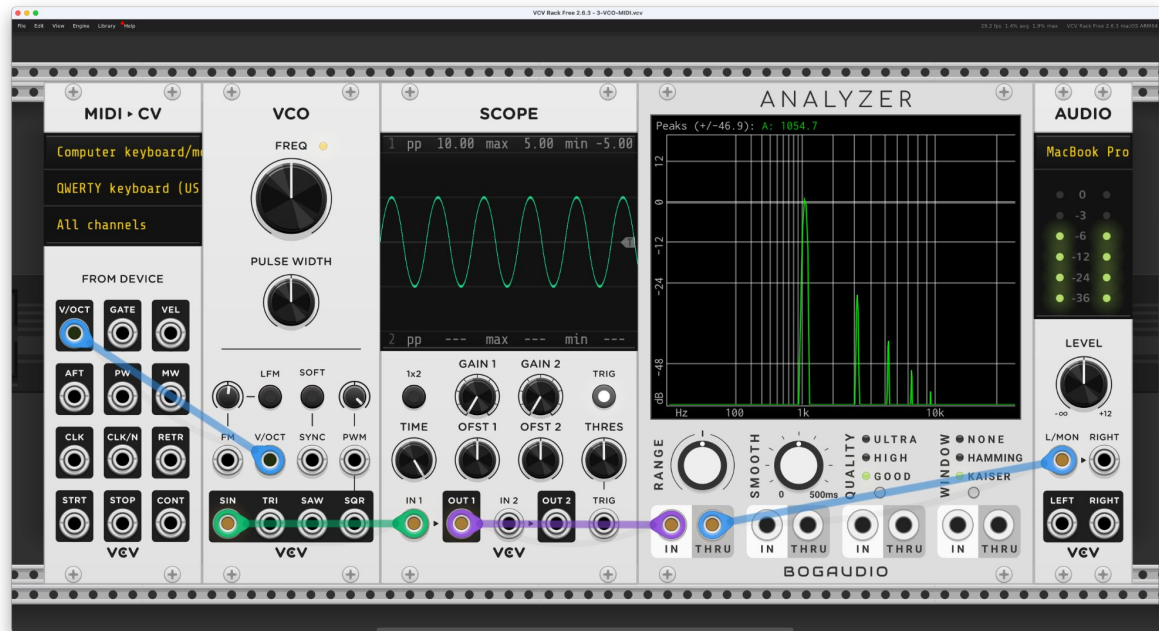
MIDI+CV

3-VCO-MIDI.vcv

Can use computer keyboard or MIDI controller if you have one to control pitch

If you have your Arduino sequencer still built - you could connect it!

No “gating” yet...



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Envelopes

4-VCO-ADSR-VCA.vcv

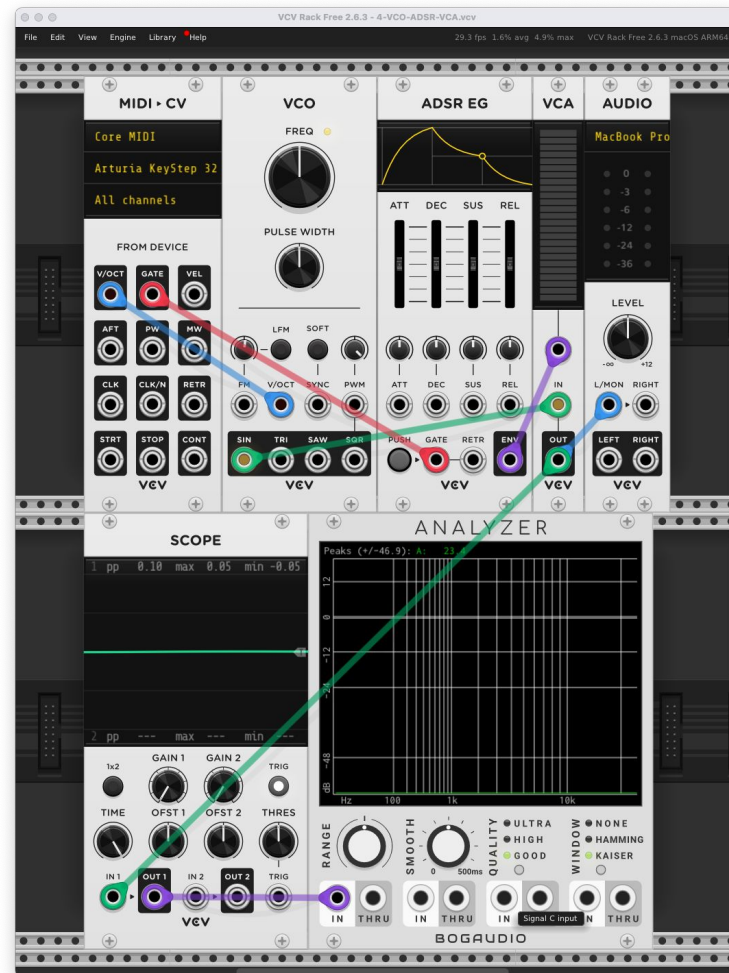
ADSR EG generates the envelope shape

Adjust ATT/DEC/SUS/REL

VCA amplifies (multiplies) IN signal by the CV input

Voltage controlled amplifier makes the signal louder or softer - across all frequencies

By triggering the envelope from the GATE we have control over note timing



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Voltage Controlled Filter

5-VCO-VCF-ADSR-VCA.vcv

Filters amplify different frequencies by different amounts

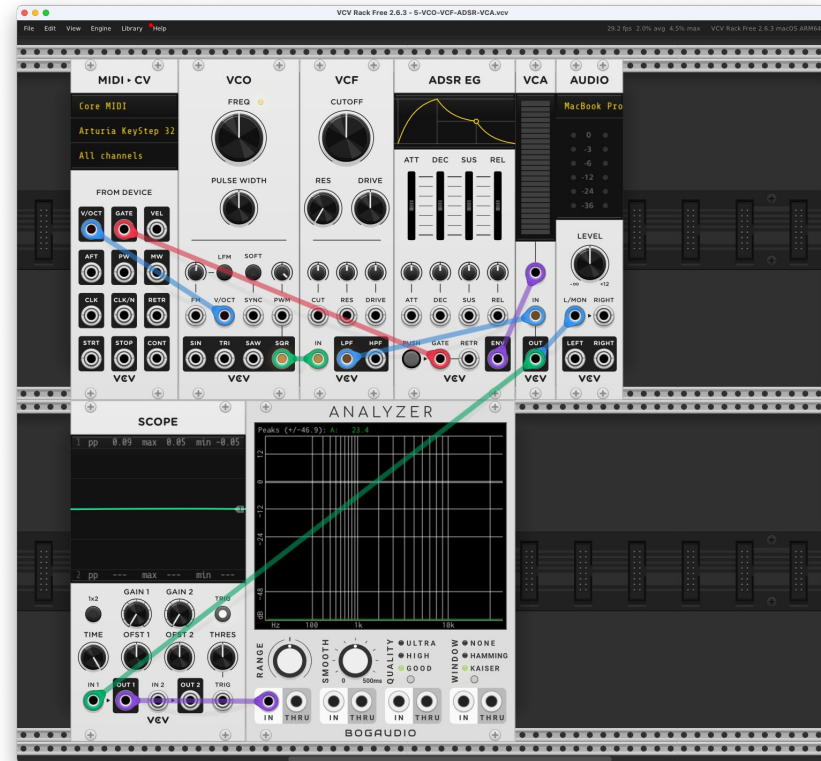
Low pass: Attenuate (make softer) frequencies above the cutoff

High pass: Attenuate frequencies below the cutoff

Patch the square wave from the VCO (harmonically rich)

Adjust cutoff & observe ANALYZER - low cutoff is similar to SINE

Try the HPF



Filter Envelope

6-Filter-Envelope.vcv

Use the ADSR EG to not only control VCA level, but also the VCF cutoff frequency

Need to adjust CUT CV knob (controls how much of the incoming CV affects the CUTOFF value)

The ADSR EG is automatically twisting the CUTOFF knob

Adjust the big CUTOFF knob to vary the effect

Could use a separate ADSR EG to control the filter cutoff

7-Filter-Separate-Envelope.vcv



Sequencers

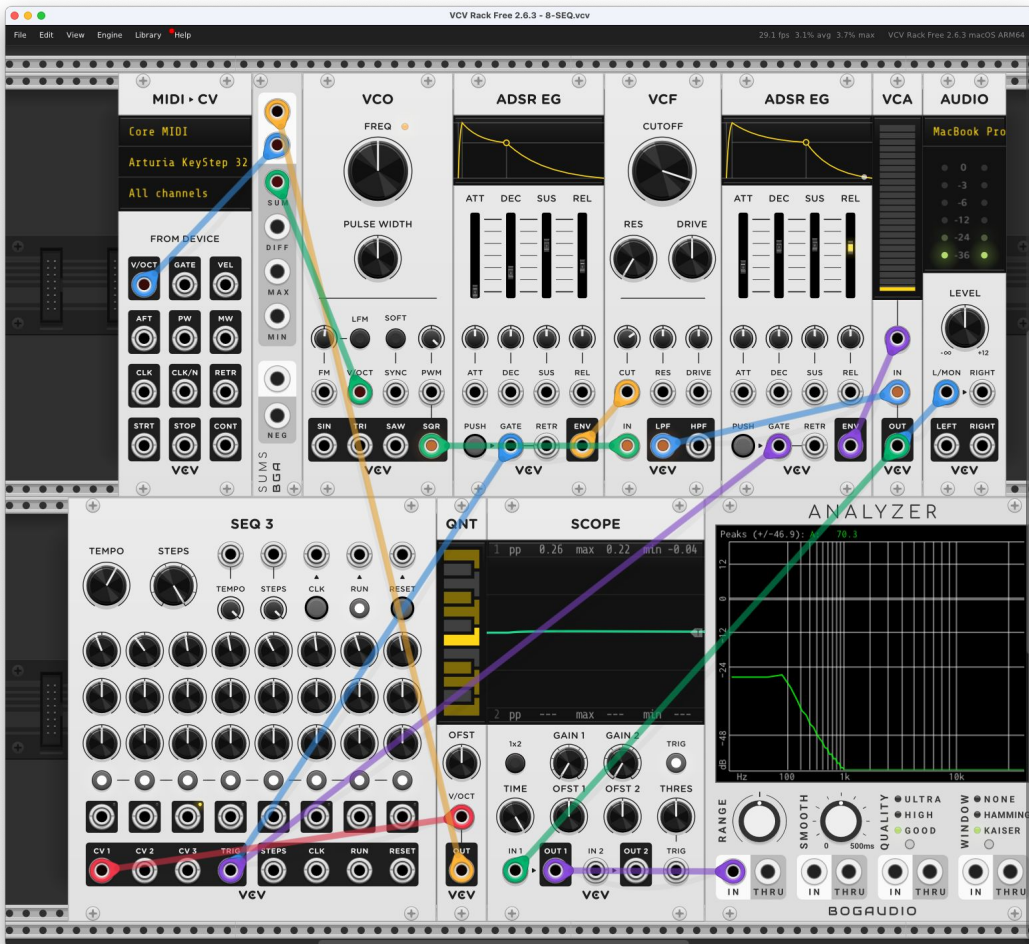
8-SEQ.vcv

SEQ 3 provides a sequence of voltages set by the CV1 knobs (top row)

These are passed to **QNT** which quantizes to a musical scale (configure by Right click->Preset)

SEQ 3 TRIG now triggers the **GATE** for both envelopes

SUMS is used to add the V/OCT outputs from **MIDI** and **QNT** this allows you to transpose the sequence



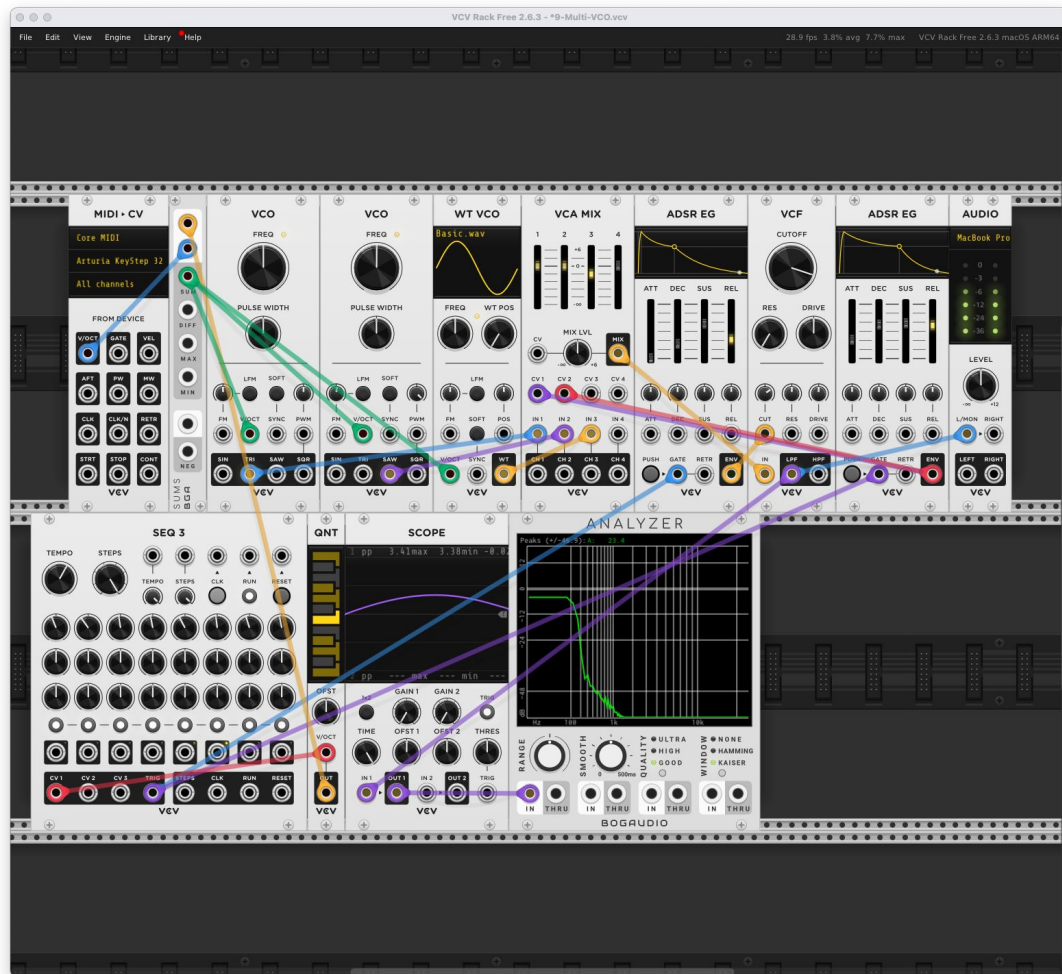
More complex waveforms

9-Multi-VCO.vcv

Add extra VCOs and mix together

Try detuning one of the VCOs

Adjust the WT POS



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Low Frequency Oscillators

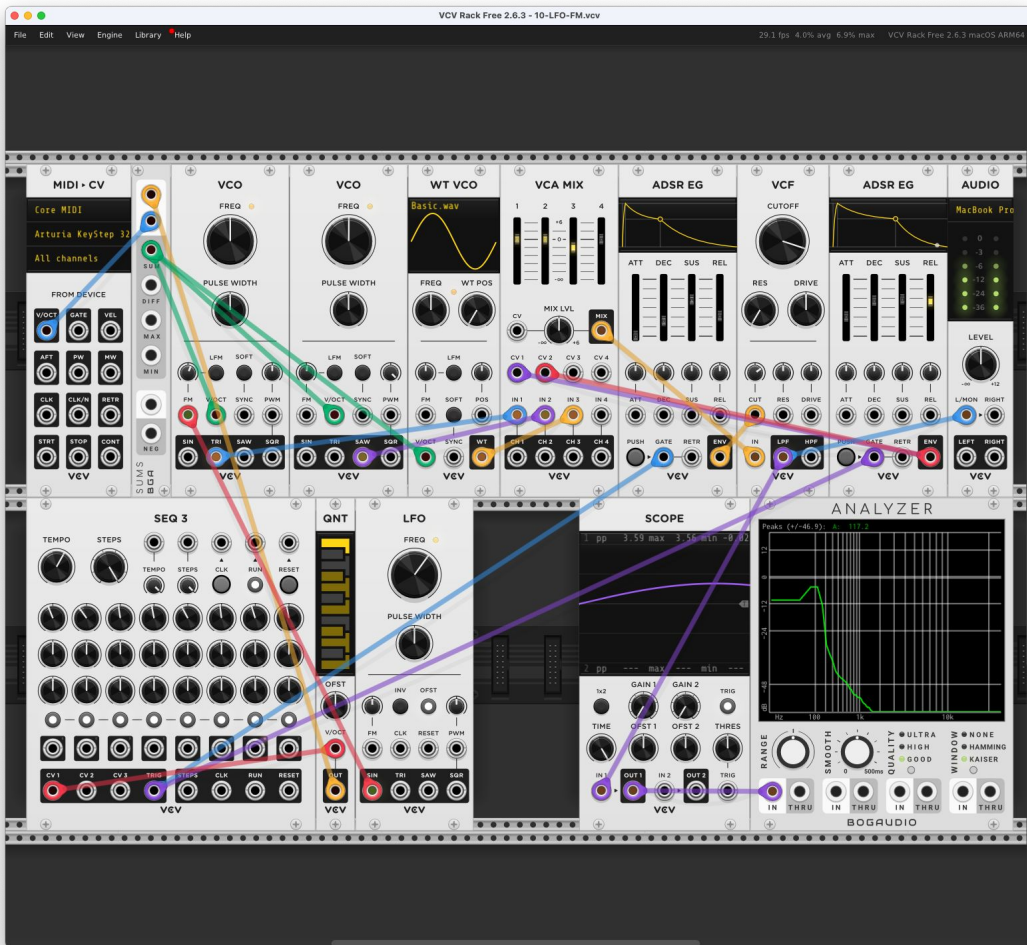
10-LFO-FM.vcv

Just like oscillators, just lower frequency

For “twiddling knobs”

Try changing the LFO rate or wave

Can modulate any CV...



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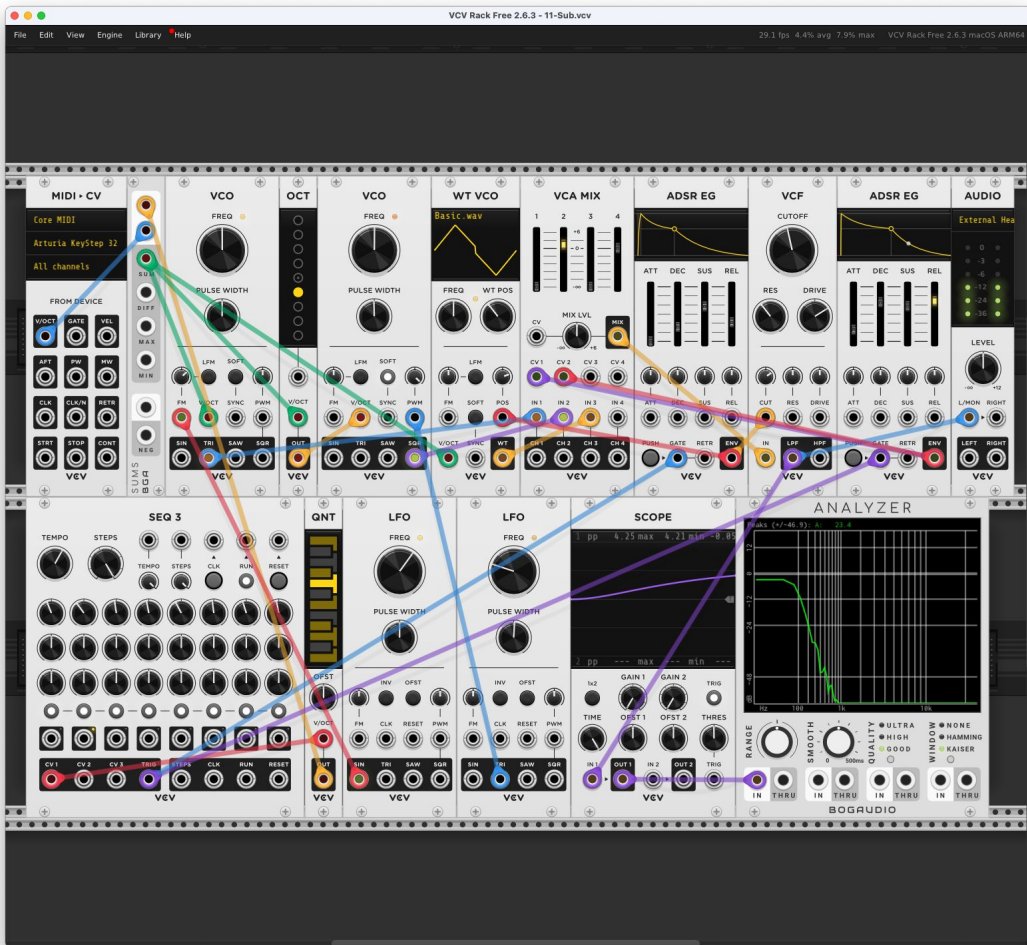
Modulation

11-SUB.vcv

OCT module translates the V/OCT by whole octaves

Using an LFO to sweep the pulse width of VCO 2

Using EG to ramp the wavetable



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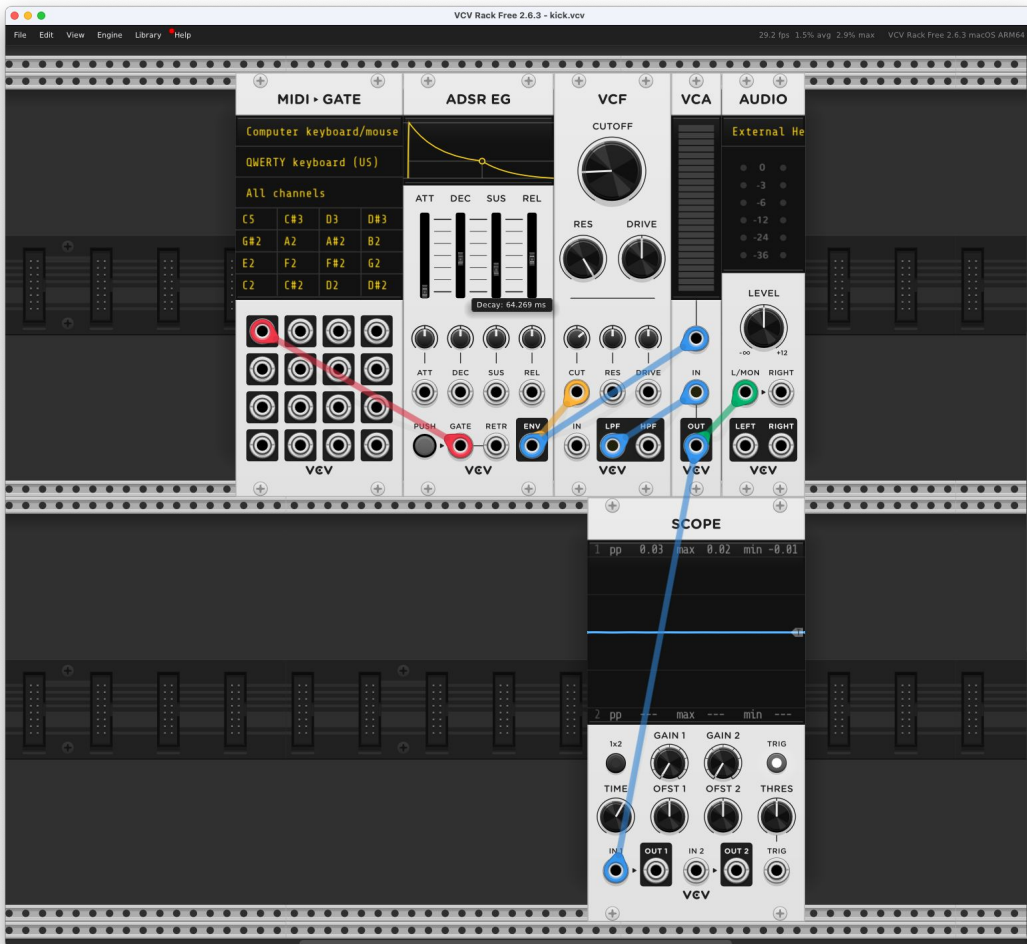
Kick Drum

Crank resonance to self-oscillate the VCF

Cutoff frequency sets the tone

Use ADSR to provide a decay profile to the cutoff

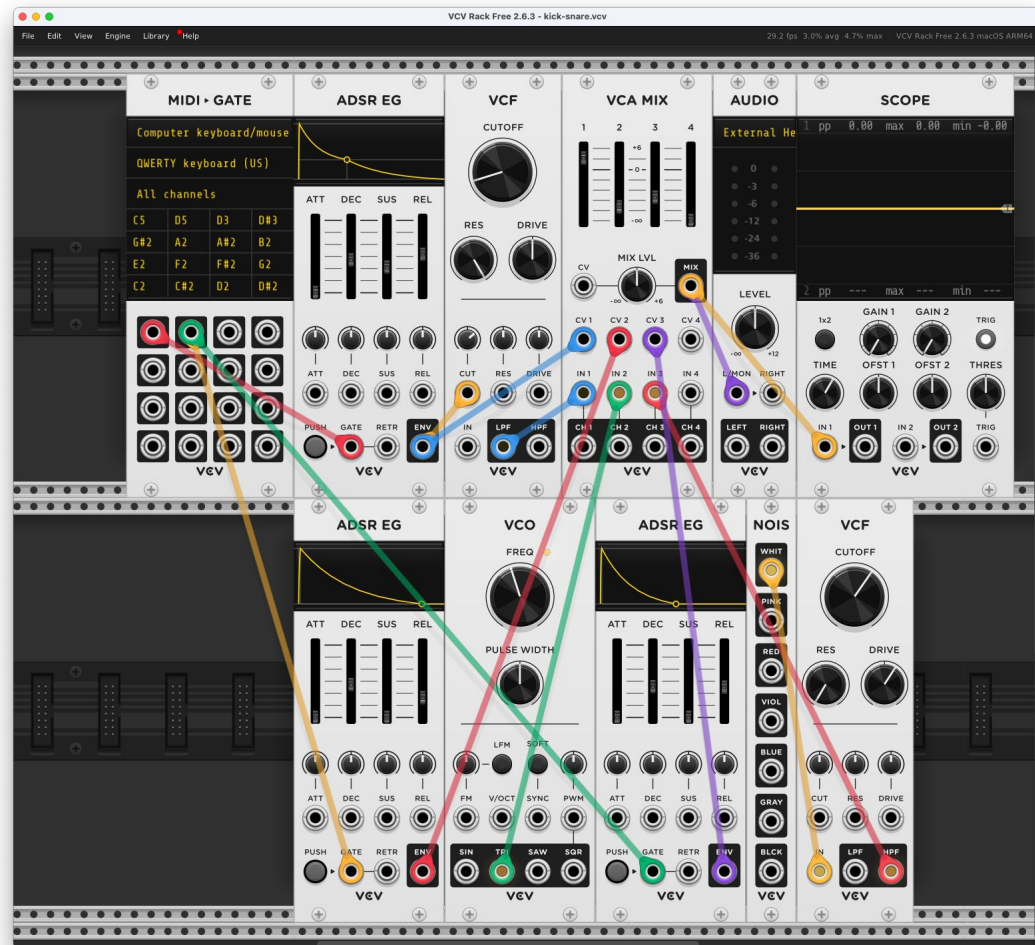
Adjust envelope/cutoff/resonance to tweak sound



Snare Drum

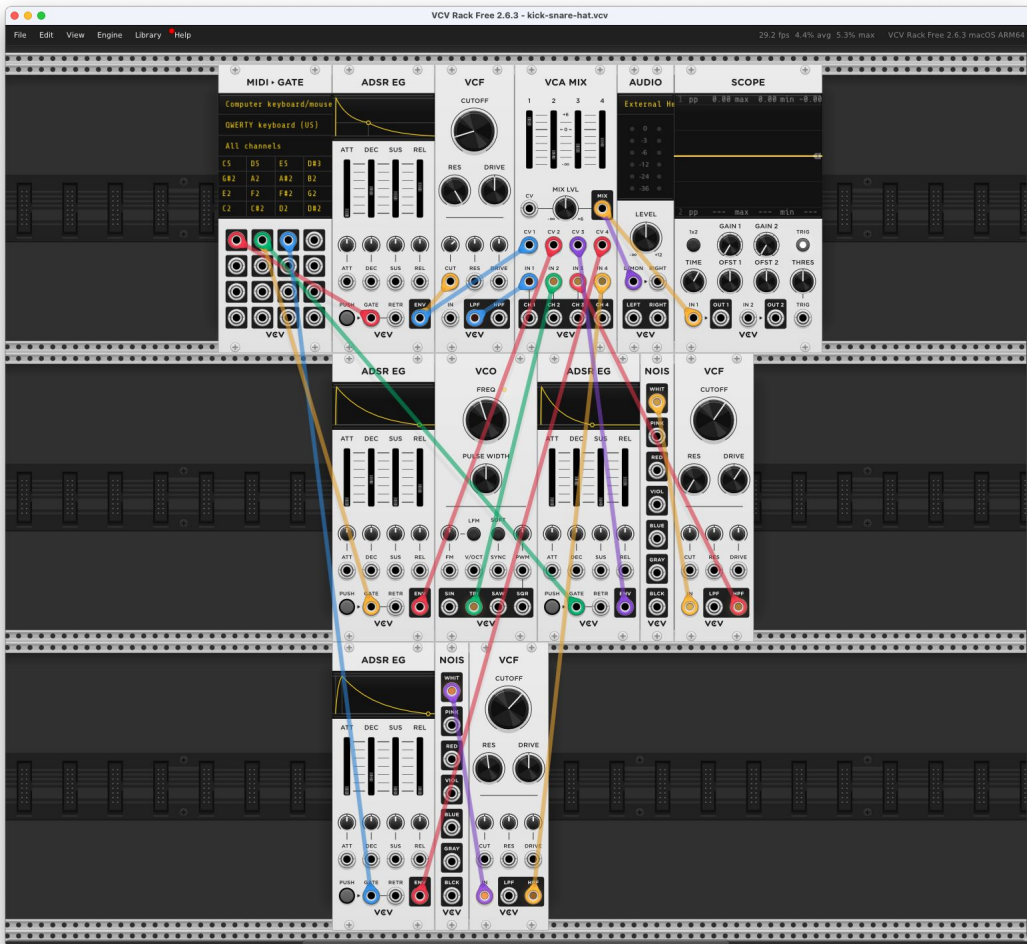
Layer a SIN OR TRI waveform with high pass filtered noise

Use short decay envelopes



Hihat

High pass filtered noise with short decay



Next Week...

Outline

We will be building a homebrew MIDI controller using various sensors

This can be used to control a DAW (e.g. Ableton or Garageband), or VCV Rack

Experiment with VCV Rack during the week to design some patches...

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